

■ KITE LMS — Enterprise User Guide

Complete Operational Manual & Interconnected Multi-Tenant Architecture Guide

System Overview: KITE LMS is a multi-tenant Learning Management System designed for educational institutes and enterprise training centers. The application segregates access across **4 primary login portals** (Platform Admin, Institute Admin, Teacher, Student) while using a unified data model where CRM leads, course delivery, assessments, billing, and automated workflows seamlessly interconnect.

1. Primary Login Portals & Access Scopes

KITE LMS partitions responsibilities into 4 dedicated web portals, each customized for specific administrative, teaching, or learning activities. All portals share single sign-on (SSO) role authentication with strict tenant isolation.

Portal Role	Primary Login Route	Core Responsibilities & Capabilities
■■ Platform Admin	/admin/login	SaaS ecosystem monitoring, tenant institute provisioning, custom domain assignments, and global add-on feature toggles.
■ Institute Admin	/institute-admin/login	Curriculum setup, CRM lead pipeline & 1-click enrollment, automated workflow recipes, financial GST billing, and certificate templates.
■■■ Teacher / Instructor	/teacher/login (or /login)	Assigned course delivery, live virtual class hosting (Zoom/Meet), student attendance tracking, quiz building, and assignment solution file evaluation.
■ Student / Learner	/student/login	Enrolled course dashboard, instant 1-click live class join, video module streaming, timed quiz attempts, task submission, and certificate downloads.

2. End-to-End Interconnected Data Lifecycle

A central strength of KITE LMS is how data flows automatically between modules. The sequence below demonstrates how a prospective lead moves through CRM, automated workflows, course delivery, assessments, invoicing, and certification:

Lifecycle Step	Detailed Functional Interconnection
Step 1: Lead Capture (CRM)	Inquiry captured via CRM Lead form with student name, contact details, and course interest.
Step 2: Follow-up & One-Click Enrollment	Institute Admin or Teacher logs call follow-ups. Clicking 'Enroll Student' converts Lead to a Student record and assigns target Course.
Step 3: Workflow Trigger Dispatch	System fires 'Student Enrolled' and 'Course Access Granted' events. WorkflowEngine evaluates recipes and executes configured actions (e.g. welcome email).
Step 4: Invoice & Payment Transaction	Billing Engine computes GST tax (subtotal + tax_amount), generates sequential invoice number (e.g. KITE-INV-000001), and logs successful transaction.
Step 5: Student Learning & Live Classes	Student accesses student portal to join scheduled Zoom/Meet live classes. Teacher conducts session and marks attendance (Present/Absent/Late).

Step 6: Quizzes & Task Evaluation	Student submits timed interactive quiz (auto-scored via QuizAnswer keys) and uploads solution files for tasks. Teacher reviews and grades submission.
Step 7: Automated Certificate Generation	Upon course completion, system verifies student eligibility and generates verified certificate record (CERT-YYYYMMDD-XXXX) downloadable as PDF.

3. Detailed Module Operational Breakdown

3.1 Curriculum & Course Builder

Institutes structure learning material using a 4-tier hierarchy: **Course** → **Batch** → **Prerecorded Module** → **Lecture**.

- **Courses:** Defines title, code, description, visibility (Public/Private), and status (Draft/Published).
- **Batches:** Segregates students into active cohort groups (e.g. 'Batch A 2026') for targeted class scheduling.
- **Prerecorded Modules & Lectures:** Organizes recorded video lessons with custom ordering and video streaming URLs.

3.2 Live Classes & Interactive Virtual Meetings

Teachers schedule interactive live classes for assigned courses/batches using Zoom, Google Meet, or custom providers.

- **1-Click Join:** Students see today's live classes on their dashboard with instant 'Join Meeting' buttons.
- **Attendance Tracking:** Teachers mark attendance (Present, Absent, Late) which updates student performance metrics.
- **Session Recordings:** Post-class recording URLs are attached to the course portal for student revision.

3.3 Quizzes, Assessments & Task Submissions

KITE LMS supports both automated objective evaluation and manual assignment grading.

- **Interactive Quizzes:** Supports timed multiple-choice questions. Submissions calculate immediate score cards and record QuizAnswer option selections.
- **Task Assignments:** Teachers post tasks with instructions and file attachments. Students upload solution files before deadlines for teacher review and feedback.

3.4 CRM Lead Pipeline & Automated Workflows

Captures prospective inquiries and automates institute administrative recipes.

- **CRM Lead Pipeline:** Tracks lead stages (New, Contacted, Follow-up, Converted). One-click enrollment creates student user accounts instantly.
- **Automated Workflow Engine:** Configures event-condition-action recipes (e.g. *If Payment Received > \$0 → Grant Course Access & Send Email*) with WorkflowExecutionLog tracking.

3.5 Financial Suite, GST Billing & Certificate Engine

Enterprise revenue management and automated credential verification.

- **GST Billing & Invoices:** Configures tax percentage (e.g. 18%), invoice prefixes (e.g. KITE-INV-), auto-incrementing numbers, and generates formal invoice records.
- **Verified Certificates:** Custom certificate templates issue official completion records with unique reference IDs (CERT-YYYYMMDD-XXXX).

4. Default Test Credentials & Quick Reference

After initializing the database via python seed.py, the following default credentials are operational across all 4 login portals:

Portal Role	Username / Email	Password	Primary Access Scope
Platform Admin	<code>platformadmin@kite.lms</code>	password123	Multi-Tenant Management & SaaS Add-ons
Institute Admin	<code>instituteadmin@kite.lms</code>	password123	Curriculum, CRM, Workflows, Billing & Certificates
Teacher	<code>teacher@kite.lms</code>	password123	Assigned Courses, Live Classes, Quizzes & Attendance
Student	<code>student@kite.lms</code>	password123	Learning Dashboard, Live Join, Tasks & Certificates

Operation	Command	URL / Endpoint
Start Backend	<code>uvicorn app.main:app --reload</code>	<code>http://localhost:8000/docs</code>
Start Frontend	<code>npm run dev</code>	<code>http://localhost:5173</code>
Seed Database	<code>python seed.py</code>	Local Database File (test.db)
Frontend Build	<code>npm run build</code>	Production Bundle (dist/)